SOLUTIONS: DESIGN OF LINEAR MULTIVARIABLE CONTROL SYSTEMS

1. a) Let $\Delta(s)$ and $H(s)$ be stable. The feedback loop is stable if $\|\Delta H\|_\infty < 1$.

b) Let the input to Δ be ε while the output be δ. Then a calculation shows that $\varepsilon = -S\delta$ where $S = (I+GK)^{-1}$ is the sensitivity. Using the small gain theorem, to satisfy the robustness design requirement, it is sufficient that $\|\Delta(j\omega)S(j\omega)\| < 1, \forall\omega$. This can be satisfied if $\|WS\|_\infty < 1$, where $W = wI$.

A calculation shows that, when $\Delta = 0$,

$$u(s) = K(s)S(s)r(s).$$

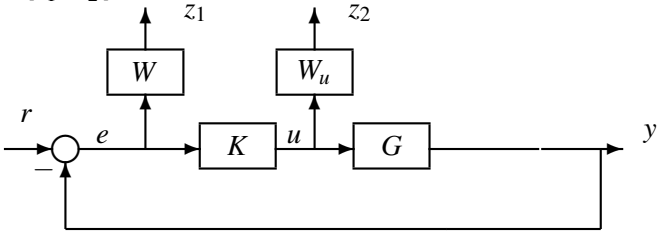
Thus

$$\|u(j\omega)\| \le \|K(j\omega)S(j\omega)\|\|r(j\omega)\|,$$

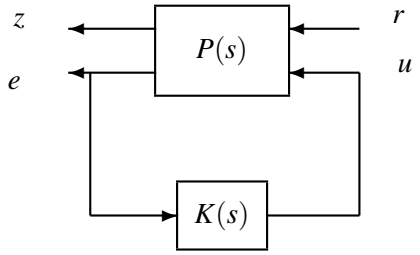
It follows that a sufficient condition to achieve the performance design specification is $\|K(j\omega)S(j\omega)\| < |w_u^{-1}(j\omega)|$ for all ω or equivalently $\|W_u KS\|_\infty < 1$, where $W_u = w_u I$.

To satisfy all design requirements, it is sufficient that $\left\| \begin{bmatrix} WS \\ W_u KS \end{bmatrix} \right\|_\infty < 1$.

c) So, the design specifications reduce to the requirement that the transfer matrix from r to $z = [z_1^T \;\; z_2^T]^T$ in the following diagram has $\mathscr{H}_\infty$-norm less than 1.



d) The generalized regulator problem: find an internally stabilizing K such that $\|T_{zr}\|_\infty < 1$



$$P = \begin{bmatrix} P_{11} & P_{12} \\ P_{21} & P_{22} \end{bmatrix} = \left[\begin{array}{c|c} -W & -WG \\ 0 & W_u \\ \hline -I & -G \end{array}\right].$$

2. a) i) Assume first that A is stable. Then $\|H\|_\infty < \gamma$ if and only if, with $x(0) = 0$, $J := \int_0^\infty [y^T y - \gamma^2 u^T u]dt < 0$, for all $u(t)$ such that $\|u\|_2 < \infty$. If $\|u\|_2$ is bounded, then $\lim_{t\to\infty} x(t) = 0$. Now, $\int_0^\infty \frac{\mathrm{d}}{\mathrm{dt}}[x^T Px]dt = x(\infty)^T Px(\infty) - x(0)^T Px(0) = 0$. So,

$$0 = \int_0^\infty \left(\dot{x}^T Px + x^T P\dot{x}\right) dt = \int_0^\infty [x^T (A^T P + PA)x + x^T PBu + u^T B^T Px]dt.$$

Use $y = Cx + Du$ and add the last expression to J

$$\begin{aligned} J &= \int_0^\infty [x^T (A^T P + PA + C^T C)x + 2x^T (PB + C^T D)u + u^T (D^T D - \gamma^2 I)u]dt \\ &= \int_0^\infty \begin{bmatrix} x^T & u^T \end{bmatrix} \overbrace{\begin{bmatrix} A^T P + PA + C^T C & PB + C^T D \\ B^T P + D^T C & D^T D - \gamma^2 I \end{bmatrix}}^{M} \begin{bmatrix} x \\ u \end{bmatrix} dt. \end{aligned}$$

It follows that $J < 0$, and so $\|H\|_\infty < \gamma$, if $P \succ 0$ and $M \prec 0$. It remains to prove the stability of A. Note that from the (1,1) entry of M we have $A^T P + PA \prec 0$. Let $Az = \lambda z$, $z \neq 0$. Then multiplying the inequality from the left by z^T and from the right by z gives $(\lambda + \bar{\lambda})z^T Pz < 0$. Since $P \succ 0$ it follows that $z^T Pz > 0$ and it follows that $\lambda + \bar{\lambda} < 0$ so that A is stable.

ii) Now, $H(s)^T = D^T + B^T (sI - A^T)^{-1} C^T$. It follows that the dual conditions are: there exists a $P = P^T \succ 0$ such that

$$\begin{bmatrix} AP + PA^T + BB^T & PC^T + BD^T \\ CP + DB^T & DD^T - \gamma^2 I \end{bmatrix} \prec 0.$$

b) i) The state space equations are

$$\begin{aligned} \dot{x}(t) &= Ax(t) - LCx(t) + Lr(t) = (A - LC)x(t) + Lr(t) \\ y(t) &= Cx(t) \end{aligned}$$

Therefore $T_{yr}(s) = C\left(sI - (A - LC)\right)^{-1} L$.

ii) Using the answer to Part a(i), replacing A by $A - LC$ and setting $D = 0$ and $B = L$, there exists a feasible matrix L if there exists $P \succ 0$ such that

$$\begin{bmatrix} (A - LC)^T P + P(A - LC) + C^T C & PL \\ L^T P & -\gamma^2 I \end{bmatrix} \prec 0.$$

Since the only nonlinearity is due to the product PL, we let $Y = PL$ and so there exists a feasible gain matrix L if there exists $P \succ 0$ and Y such that

$$\begin{bmatrix} A^T P + PA - YC - C^T Y^T + C^T C & Y \\ Y^T & -\gamma^2 I \end{bmatrix} \prec 0.$$

3. a) Let $V = x'Px$ and set $u = -Fx$. Provided that $P = P' \succ 0$ and $\dot{V} < 0$ along closed-loop trajectories, we can assume $\lim_{t\to\infty} x(t) = 0$. Then

$$\dot{V} = \dot{x}'Px + x'P\dot{x} = x'\left(A'P + PA - F'B'P - PBF\right)x.$$

Integrating from 0 to ∞ and using $x(\infty) = 0$,

$$\int_0^\infty x'\left(A'P + PA - F'B'P - PBF\right)xdt = -x_0'Px_0.$$

b) Using the definition of J, adding the last equation and completing a square:

$$J = x_0'Px_0 + \int_0^\infty \{x'[A'P + PA + I - PBB'P]x + \|(F - B'P)x\|^2\}dt.$$

Since the last term is always nonnegative, it follows that the minimizing value of F is given by $F = B'P$. We can set the term in square brackets to zero provided P satisfies the Riccati equation,

$$A'P + PA + I - PBB'P = 0.$$

It follows that the minimum value of J is $x_0'Px_0$.

c) We need to prove that $A_c := A - BB'P$ is stable. The Riccati equation can be written as $A_c'P + PA_c + I + PBB'P = 0$. Let $\lambda \in \mathscr{C}$ be an eigenvalue of A_c and $z \neq 0$ be the corresponding eigenvector. Pre– and post–multiplying the Riccati equation be z' and z respectively gives $(\lambda + \bar{\lambda})z'Pz + z'z + z'PBB'Pz = 0$. Since $P \succ 0$ and $z \neq 0$, $z'Pz > 0$, $z'z > 0$ and $z'PBB'Pz \geq 0$. It follows that $\lambda + \bar{\lambda} < 0$ and the closed loop is stable.

d) The loop-gain is given by $L(s) = F(sI - A)^{-1}B$. Let λ be a zero of $L(s)$ so that,

$$\begin{bmatrix} A - \lambda I & B \\ F & 0 \end{bmatrix}\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix},$$

for some $v_1,\ v_2$ with $v_1 \neq 0$ since B has full column rank. Then,

$$Av_1 = \lambda v_1 - Bv_2,$$

and,

$$Fv_1 = 0 \Rightarrow B'Pv_1 = 0.$$

Consider the Riccati equation. Then,

$$\begin{aligned} 0 &= v_1'[A'P + PA - PBB'P + I]v_1 \\ &= (\bar{\lambda}v_1' - v_2'B')Pv_1 + v_1'P(\lambda v_1 - Bv_2) + v_1'v_1 \\ &= (\bar{\lambda} + \lambda)\underbrace{v_1'Pv_1}_{>0} + \underbrace{v_1'v_1}_{>0} \end{aligned}$$

Hence $\lambda + \bar{\lambda} < 0$ and $L(s)$ is minimum-phase.

4. a) A suitable Lyapunov function is $V = x^T Px$ where $P = P^T$. Provided that $P = P^T \succ 0$ and $\dot{V} < 0$ along closed-loop trajectories, we have $\lim_{t\to\infty} x(t) = 0$. Then

$$\dot{V} = \dot{x}^T Px + x^T P\dot{x} = x^T \left(A^T P + PA - F^T B_2^T P - PB_2 F\right) x.$$

Hence $\int_0^\infty \dot{V}(t)dt = \int_0^\infty x^T \left(A^T P + PA - F^T B_2^T P - PB_2 F\right) xdt = -x_0^T Px_0$.

Adding the last equation to J_0 and completing a square:

$$J_0 = x_0^T Px_0 + \int_0^\infty \{x^T [A^T P + PA + C_1^T C_1 - PB_2 B_2^T P]x + \left\|(F + B_2^T P)x\right\|^2\} dt.$$

Since the last term is always nonnegative, it follows that the minimizing value of F is given by $F = -B_2^T P$. We can set the term in square brackets to zero provided P satisfies the Riccati equation $A^T P + PA + C_1^T C_1 - PB_2 B_2^T P = 0$. Finally, we prove that $A_c := A - B_2 B_2' P$ is stable. The Riccati equation can be written as $A_c' P + PA_c + C_1^T C_1 + PB_2 B_2' P = 0$. Let $\lambda \in \mathscr{C}$ be an eigenvalue of A_c and $y \neq 0$ be the eigenvector. Then $(\lambda + \bar{\lambda}) y'Py + y'C_1' C_1 y + y'PB_2 B_2' Py = 0$. Since $P \succ 0$ and $y \neq 0$, $y'Py > 0$, $y'C_1' C_1 y \geq 0$ and $y'PB_2 B_2' Py \geq 0$. It follows that $\lambda + \bar{\lambda} \leq 0$ with $\lambda + \bar{\lambda} = 0$ if and only if $B_2' Py = 0$ and $C_1 y = 0$. But this implies $Ay = \lambda y$ and $C_1 y = 0$, which contradicts the observability of the pair (A, C_1). This proves that $\lambda + \bar{\lambda} < 0$ and so the closed loop is stable.

b) i) Let $J = \|z\|_2^2 - \gamma^2 \|w\|_2^2$ denote the cost function. Let $V = x'Xx$ and set $u = Fx$. Provided that $X = X' \succ 0$ and $\dot{V} < 0$ along the closed-loop trajectory, we can assume $\lim_{t\to\infty} x(t) = 0$. Then

$$\dot{V} = \dot{x}'Xx + x'X\dot{x} = x'\left(A'X + XA + F'B_2'X + XB_2F\right)x + x'XB_1w_1 + w_1'B_1'Xx.$$

Integrating from 0 to ∞ and using $x(\infty) = x_0 = 0$,

$$0 = \int_0^\infty [x'\left(A'X + XA + F'B_2'X + XB_2F\right)x + x'XB_1w_1 + w_1'B_1'Xx]dt.$$

Adding the last equation to the definition of J and completing squares

$$J = \int_0^\infty \{x'\left[A'X + XA + C_1'C_1 - X\left(B_2B_2' - \gamma^{-2}B_1B_1'\right)X\right]x + \|Zx\|^2 - \|v\|^2\} dt$$

where $Z = F + B_2'X$ and $v = \gamma w - \gamma^{-1}B_1'Xx$.

ii) The requirement $\|H\|_\infty \leq \gamma$ is equivalent to $J \leq 0$ when $x_0 = 0$. Thus sufficient conditions for $J \leq 0$ are $Z = 0$ and the existence of X s.t.

$$A'X + XA + C_1'C_1 - X\left(B_2B_2' - \gamma^{-2}B_1B_1'\right)X = 0, \qquad X = X' \succ 0.$$

Setting $Z = 0$, the control policy is $F = -B_2'X$. Setting $v = 0$, the worst case disturbance is $w^* = \gamma^{-2}B_1'X$. The closed-loop with $u = Fx$ and $w = w^*$ is $\dot{x} = A_c x$ where $A_c = A - \left(B_2B_2' - \gamma^{-2}B_1B_1'\right)$ and a further condition is $Re\ \lambda_i(A_c) < 0,\ \forall\ i$.

c) By letting $\gamma \to \infty$ in part (b), it is noticed that the solution becomes identical to that of part (a). It follows that we can vary γ in part (b) as tradeoff between maximum performance ($\gamma \to \infty$) and maximum robustness ($\gamma \to \gamma_o$, where γ_o is the smallest γ satisfying the requirements of part (b)).

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